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Smart Polymers for Sensing Technologies

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Chapter 3
Toward Responsive Polymer Actuators Based on Novel Composite Structures



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Abstract Recent advancements in materials science have highlighted the growing demand for materials with complex properties, such as flexibility, biodegradability, and multifunctionality, driving the development of smart materials. These materials are typically functional polymers that respond to various external stimuli, including temperature, electric fields, and magnetic fields. Among devices derived from smart materials, smart actuators have attracted significant attention for their broad applications in biomedical devices, soft robotics, and wearable electronics. Smart actuators can be easily customized through chemical or physical strategies to perform different tasks, often surpassing traditional devices in performance. One such strategy is the development of smart polymer composites. This research briefly explores recent advancements in smart polymer composites used as actuators. Based on their type of responsiveness, smart composite actuators can be categorized into several classes. Herein, three well-known categories are discussed: electroactive polymers, thermo-responsive polymers, and magneto-responsive polymers. Special emphasis is placed on responsive hydrogels and shape memory polymers. Additionally, other actuator types—such as dielectric elastomers, ionic polymer-metal composites, and magnetorheological elastomers—are introduced. By showcasing advancements in polymer composite actuators across various fields, this study underscores the importance of developing emerging devices and technologies in smart polymer composites.

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3.1 Introduction

Polymers are macromolecules composed of repeating units called monomers, usually derived from organic compounds. Besides being cost-effective, their key advantage over metals and ceramics lies in their greater versatility. Such characteristics make polymer-based materials excellent candidates for designing state-of-the-art functionalized polymer devices, forming a significant part of the smart materials literature (see Fig. 3.1). In the context of functional polymer materials, functionality is defined as the ability to respond to various external stimuli—including thermal and pH responsiveness—in a controlled and programmable manner, which may lead to customized and unconventional applications of the resulting material [1]. Characteristics such as self-healing and biocompatibility can also be regarded as forms of functionality since they are triggered by external stimuli. Functionality can be introduced into polymer compounds through chemical modification of their composition or by altering their microstructure through the addition of extra phases. Chemical modification mainly involves incorporating functional groups into the base polymer via block copolymerization, graft copolymerization, or similar techniques. Microstructural modification, on the other hand, primarily includes the formation of polymer blends, polymer composites, and related systems.

Regarding microstructural modification scenarios, polymer composites are ideal candidates for advanced technologies [2], as they have demonstrated remarkable property enhancements compared to their base materials. Polymer composites are typically characterized by two distinct phases with specific identities within their microstructure: a base polymer phase, known as the matrix, and a secondary, usually non-polymeric phase, referred to as the reinforcement, which is incorporated into the matrix (see Fig. 3.2). The reinforcement phase—comprising micro/nanofibers, micro/nanoparticles, or other fillers—and its interaction with the matrix are responsible for the enhanced properties beyond those of the individual pristine phases, as well as for introducing added functionality to the resulting composite [3]. The versatility and tunability of responsive polymer composites make them indispensable in various fields, including tissue engineering, wearable bioelectronics, soft robotics, smart textiles, etc. [4–8] (see Fig. 3.3).

This study primarily reviews the literature on functional polymer composites used as actuators. Herein, the term "actuator" is used in a broader sense, referring to any notable responsiveness of functional polymers to external stimuli. The functional polymer composites discussed herein fall into three main categories: electroactive polymer composite actuators, including dielectric elastomer actuators (DEAs) and ionic polymer–metal composites (IPMCs), which are discussed in Sect. 3.2.1; thermally responsive polymer composites, such as shape memory polymer (SMP) actuators, discussed in Sect. 3.2.2; and magneto-responsive actuators, such as magnetorheological elastomers, discussed in Sect. 3.2.3. Various types of composite hydrogels and SMPs are also discussed throughout the review, depending on their external stimulus mechanism. Furthermore, this review attempts to classify smart polymer

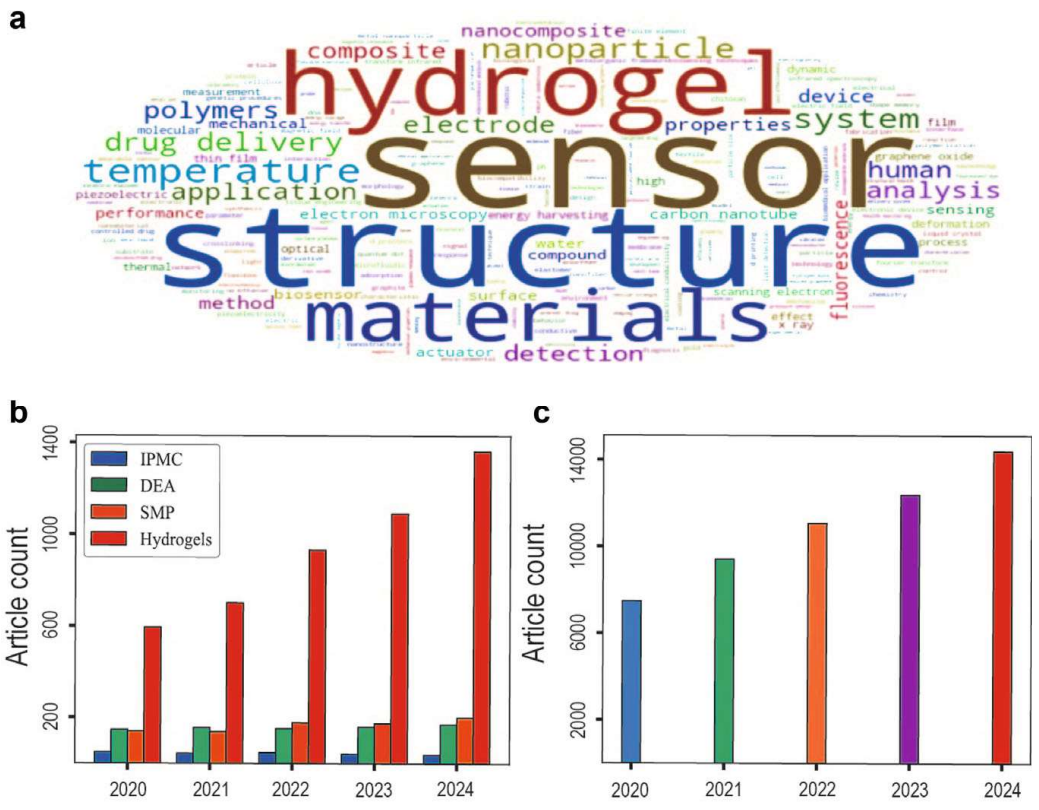


Fig. 3.1 **a** Word cloud representation of frequently used keywords in 'smart polymer actuator' literature. **b** Yearly occurrence of key terms in smart polymer actuator research. **c** Annual frequency of the 'smart polymer actuators' keyword in scientific literature. The literature database, incorporating the abstracts and titles of over 20,000 articles published between 2020 and 2024, was obtained from Scopus

actuators with architected structures as structural composites. Finally, related theoretical studies are briefly highlighted to support experimental work on smart materials, providing a comprehensive overview of the current state of the field and identifying existing research gaps.

3.2 Responsive Polymer Actuators

3.2.1 Electroactive Polymers

Electroactive polymers (EAPs) are flexible, smart materials that reversibly produce mechanical responses in response to an applied electric field. EAPs are typically compact, stretchable, and fracture-resistant and can be broadly categorized into electronic EAPs and ionic EAPs, depending on their motion mechanism. In electronic EAPs, actuation is based on electrostatic interactions, causing a redistribution of

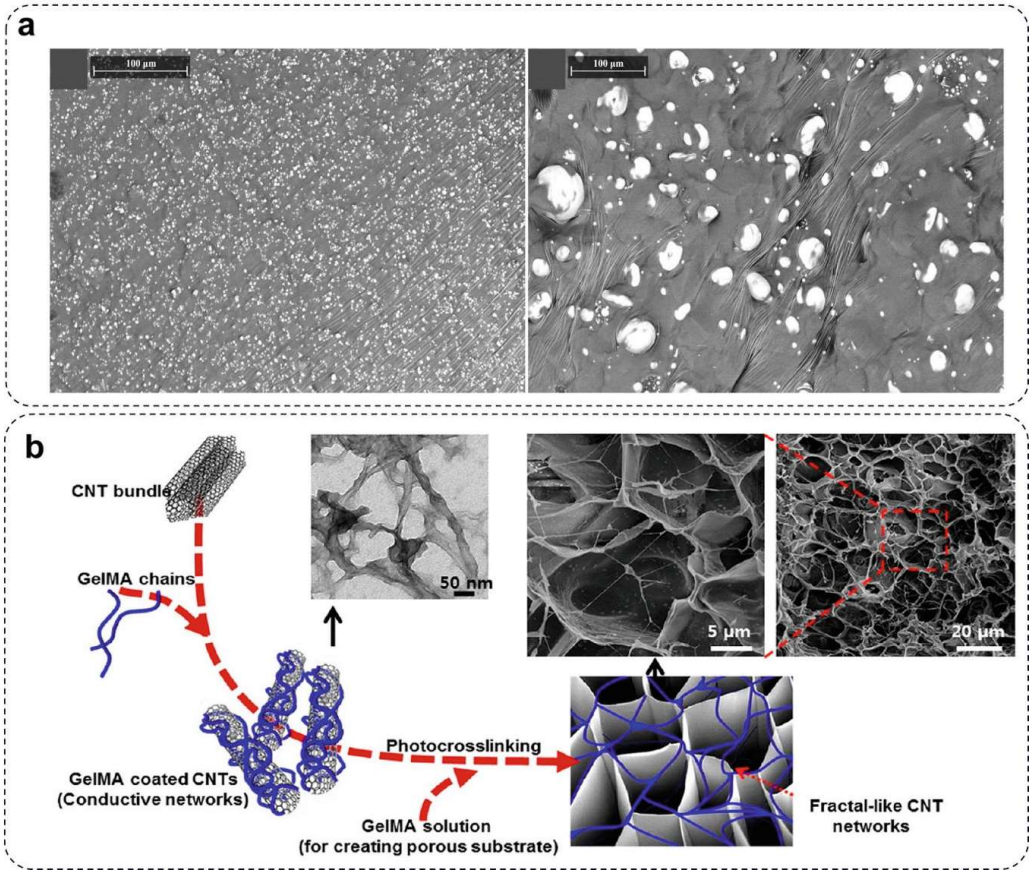


Fig. 3.2 **a** FESEM images showing the microstructural arrangement of soft and hard magnetic magnetorheological elastomers prior to magnetic actuation. Reproduced from [9] with permission. **b** Microstructure of fractal-like carbon nanotube (CNT) networks embedded in GelMA hydrogel. Nanofibrous networks are visible both across and within the porous structure. Reproduced from [10] with permission

electrons under an applied voltage, whereas in ionic EAPs, ion transport in a solvent environment causes actuation. While electronic EAPs usually have faster response times and produce higher driving forces, ionic EAPs can generate larger deformations at much lower voltages. Within this section, DEAs are introduced as electronic EAPs, followed by ionic EAPs, including ionic hydrogels and IPMCs.

3.2.1.1 Dielectric Elastomers Actuators

Dielectric elastomer actuators (DEAs) are soft actuators that utilize electrostatic forces to produce mechanical motion. They consist of an elastomeric film sandwiched between compliant electrodes. When voltage is applied to the electrodes, the elastomer layer contracts in thickness due to electrostatic forces and, as it behaves as a nearly incompressible material, expands laterally. Figure 3.4a illustrates the

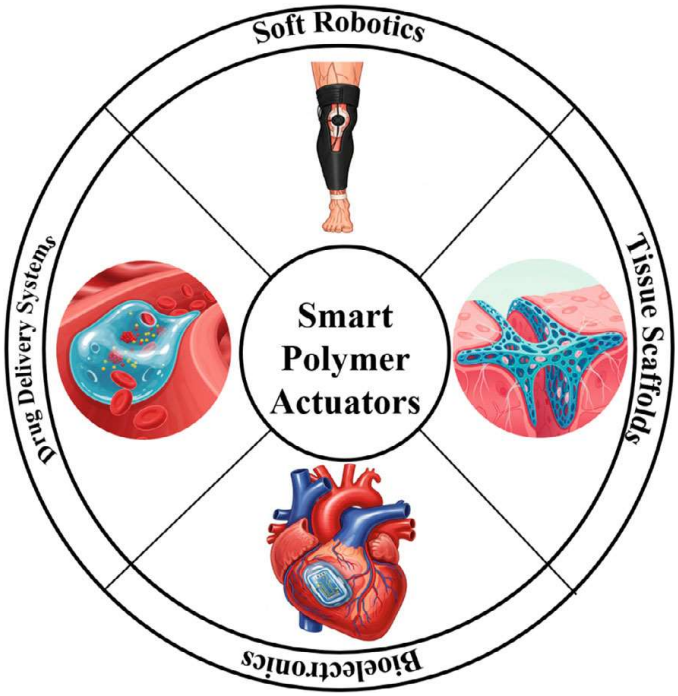


Fig. 3.3 Representative biomedical applications of smart polymer devices are illustrated, including rehabilitative soft actuators, tissue scaffolds, implantable bioelectronic systems, and stimuli-responsive gels for targeted drug delivery

actuation mechanism of a rectangular DEA when voltage is applied to its opposite sides. DEAs offer high actuation capability, fast response, cost-effectiveness, noise-free operation, and high specific energy density. These properties make DEAs suitable for various applications, including soft robotics, artificial muscles, tactile displays, and energy harvesting [11–13]. The main disadvantages of DEAs are their high driving voltages and the occurrence of electromechanical instability at relatively high electric fields.

To mitigate the above challenges, DEAs must be designed to have high dielectric constants and incorporate a strain-stiffening effect to suppress instabilities. They should also be soft while exhibiting low viscoelastic effects. Meeting all these requirements might be challenging. However, by modifying the chemical composition of the DEAs or introducing DEA composites, an overall improvement can be achieved that can simultaneously address the multiple characteristics mentioned above. In terms of DEA composites, incorporating either conductive or ceramic fillers is widely applicable. The primary approach for incorporating conductive fillers into a polymer matrix involves the sub-percolative strategy. This method maintains the dielectric nature of the matrix by working at filler concentrations below the percolation threshold while leveraging the increase in the dielectric permittivity of the composite. Carbon-based microparticles [14] and multiwall carbon nanotubes [15] have shown promise as conductive fillers. Additionally, metallic nanoparticles have shown effectiveness. For example, it has been shown that adding 4.0 vol% of aluminum nanoparticles to an acrylate copolymer considerably increases the effective dielectric constant of the composite to 8.4 [16]. Also, accepted values for the dielectric breakdown strength (140 MV/m) and maximum actuation area strain (56%)

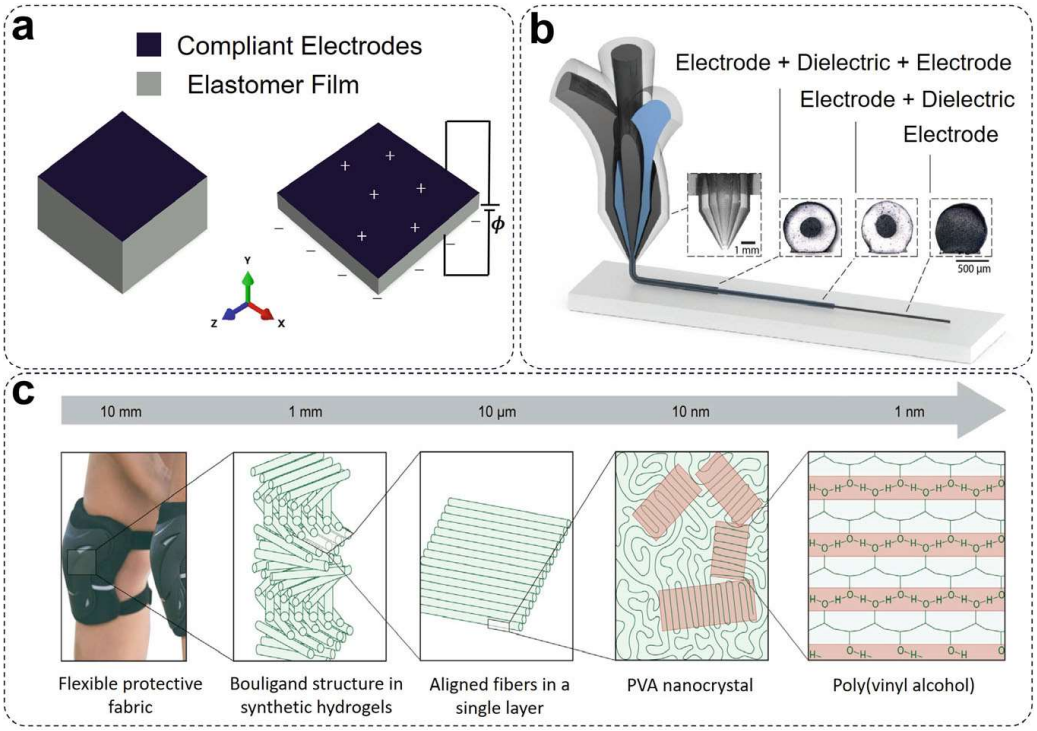


Fig. 3.4 **a** Electromechanical performance of DEAs under applied voltage (ϕ). Reproduced from [20] under the CC BY 4.0 license. **b** Multimaterial, multicore-shell 3D printing technique used for generating dielectric elastomer fibers (DEF). Reproduced from [21], with permission from John Wiley & Sons, Inc. **c** A bioinspired nanofibrous hydrogel composite with potential applications as flexible protective fabrics. Reproduced from [22], with permission from Elsevier

were observed in the samples. For ceramic fillers, various inorganic materials, such as titanium dioxide (TiO_2) [17] and calcium copper titanate ($CaCu_3Ti_4O_{12}$, or CCTO) [18], have been explored. These fillers significantly enhance the dielectric permittivity of the composite. However, researchers must carefully consider the electrical breakdown strength and dielectric loss values when introducing ceramic fillers to DEA composites. Another class of DEA composites is interpenetrating polymer networks (IPNs), which differ in structure compared to more traditional DEAs. IPNs are synthesized by diffusing and polymerizing a second polymer network into a pre-stretched one, resulting in a physically entangled structure. The main advantage of IPNs is their ability to suppress electromechanical instability due to the embedded pre-stretch within their structure while maintaining high stretchability. This approach has been applied to acrylic and silicone elastomers, demonstrating promising performance. For instance, Niu et al. [19] synthesized an acrylic-based IPN that achieved an electromechanical strain of 314% and a maximum breakdown field of 236 V/µm. Similarly, Brochu et al. demonstrated a silicone-based IPN with better actuation performance than its pre-stretched host network, achieving a linear strain of 25% and an area strain of 45%.

Apart from material improvements in DEAs, advancements in 3D printing technologies, combined with newly developed precursors possessing excellent printing

capabilities, have enabled the development of more functional DEAs with enhanced performance. The most common methods for fabricating DEAs are inkjet printing [23] and direct ink writing. These innovative techniques are paving the way for the creation of dielectric elastomer fibers and bundles with out-of-plane and programmable actuation [21], as well as self-healing DEAs with interdigitated electrodes [24], offering tunable mechanical properties and significant strain capabilities. Figure 3.4b provides a schematic of a novel technique called multimaterial, multicore-shell 3D printing, which is proposed for creating reconfigurable dielectric elastomer fibers. Beyond material enhancements, achieving controlled, reversible, and complex actuation deformations is a key area of interest in DEA research. For example, reconfigurable shape-morphing DEAs have been realized by designing multilayered DEAs with spatially varying internal electric fields [25]. Furthermore, integrating printed stiff rings into multilayer DEAs has proven effective for creating shape-morphing actuators with predefined deformations, ranging from simple conical and saddle-like forms [26] to intricate shapes like a human face [27].

In addition to experimental studies, the efficient design of DEAs necessitates an in-depth fundamental investigation of large deformation phenomena under electromechanical loading and the prediction of instability phenomena in these systems. Thus far, robust theoretical studies have been conducted to address these criteria [28–30]. Current analytical studies are moving toward DEA composites due to the broad design space they offer. Bahreman et al. [31] examined the influence of DE composite microstructure parameters on the electromechanical stability of DE composite plates, identifying unstable domains for different material compositions, volume fractions, and the properties of DE composite phases. Similarly, Majidi et al. [32] analyzed nonlinear bending-twisting coupling in the electromechanical behavior of fiber-reinforced tubular DEAs. They developed an energy-based analytical and numerical framework to predict coupled deformations and instabilities in tubular DEAs, emphasizing the critical role of fiber orientation in enhancing composite strength. Architected composite structures such as Triply Periodic Minimal Surface (TPMS) meta-structures have also been employed in the design of DEAs [20]. The findings demonstrate high levels of controllability in actuation for TPMS-based DEAs. Also, superior actuation performance is seen for TPMS-based microstructure DEAs over random and anisotropic phase-dispersed microstructure DEAs. However, the practical realization of architected-structure-based DEAs requires more advancements in the 3D printing of elastomers—a promising yet still developing technology [33, 34].

3.2.1.2 Ionic Hydrogels

Hydrogels are a class of polymer networks capable of absorbing large amounts of water within their structure while remaining insoluble due to chemical crosslinks or physical entanglements. Moreover, hydrogels exhibit flexibility comparable to natural tissue, owing to their high water content [35]. They can be tailored to be either neutral or responsive to various physical, chemical, and thermal stimuli. Their outstanding properties and responsiveness to a wide range of stimuli have made

hydrogels excellent candidates for numerous applications, including drug delivery systems, wound dressings, and microfluidics [36, 37]. Hydrogels are categorized into various classes, such as thermo-responsive, magneto-responsive, and electroactive hydrogels, depending on their specific responsiveness. This subsection focuses specifically on ionic hydrogels.

Ionic hydrogels are classified into two major types: pH-sensitive hydrogels and conductive hydrogels. The former swells or shrinks in response to environmental pH, while the latter responds to electric fields. Incorporating additional phases into pH-sensitive hydrogel composites can enhance their biocompatibility, mechanical strength, and ion transport kinetics. Regarding biocompatibility, injectable and robust hydrogel composites with interpenetrating polymer network (IPN) structures have been developed based on methacrylated alginate, demonstrating potential as dual-purpose carriers for oral drug delivery and localized sustained protein release. The in situ encapsulation of model molecules, combined with the non-cytotoxic nature of these hydrogels, underscores their safety and effectiveness for biomedical applications, particularly in targeted drug delivery [38]. Additionally, phase-separated IPN hydrogels containing both hydrophilic and hydrophobic phases have been introduced, exhibiting high strength and low hysteresis simultaneously [39]. Due to the chemical dissimilarity between the interpenetrated networks, the hydrogel's structure separates into a water-rich phase and a water-poor phase. However, the topological entanglements of the two networks prevent complete phase separation. Fatigue-resistant hydrogel composites have also been developed by incorporating nanofibers with predefined orientations. These composites are often inspired by the structures of plants and animals. A bioinspired model of nanofibrous hydrogels [22], based on the structure of a lobster's underbelly, has demonstrated superior mechanical properties, achieving a nominal strength of 8.4 MPa, a fatigue threshold of 770 J/m2, and high impact resistance in both dry (70 ± 10 kJ/kg) and hydrated (40 ± 5 kJ/kg) states. These nanofibrous hydrogels hold significant potential for applications in lightweight protection, artificial tissue replacement, and tissue engineering scaffolds. Figure 3.4c illustrates the potential application of these nanofibrous hydrogels in protective fabrics, along with their structures across different scales. Another type of ultra-strong nanofibrous hydrogel composite, inspired by the hierarchical assembly of bamboo, was also developed [40]. This composite exhibited a tensile strength of up to 60.2 MPa. In this proposed hydrogel composite, chitosan-sodium alginate nanofibers serve as the reinforcing phase, PVA acts as the hydrogel matrix, and tannic acid is used as the interfacial crosslinker to ensure strong molecular interactions.

Beyond biocompatibility and mechanical strength, hydrogel composites also enable complex actuation mechanisms. Multilayer hydrogel composites [41] and 3D-printed hydrogel composites [42] have facilitated shape-morphing capabilities such as bending, twisting, and functional transformations. Moreover, advanced 3D printing techniques, such as two-photon polymerization [43] and direct ink writing (DIW), can overcome fabrication limitations, even at the microscale. These methods provide excellent control over stiffness and porosity while minimizing material waste, offering a versatile and efficient approach to hydrogel fabrication. These promising

features pave the way for the development of more robust and biocompatible scaffolds, self-powered wearable electronics, and customized microfluidic devices. To date, hydrogel scaffolds with architected structures [44] and TPMS-based wearable sensors made from conductive hydrogels [45] have been successfully fabricated using these technologies (see Fig. 3.5a).

In terms of conductivity, conductive fillers can be incorporated into hydrogel matrices to enhance electrical performance. For instance, Ag-hydrogel composites containing silver flakes within the matrix have been developed [46], achieving a unique combination of high and stable electrical conductivity (>300 S/cm), low Young's modulus (<10 kPa), and exceptional stretchability (up to 2.5 times their original length). Furthermore, by optimizing the dispersion of silver inclusions within polyacrylamide (PAAm)-alginate hydrogels and employing a partial dehydration step, these composites maintained conductivity even after complete rehydration. These advancements establish hydrogel composites as promising materials for bioelectronics and wearable electronics. Electrophysiological recordings

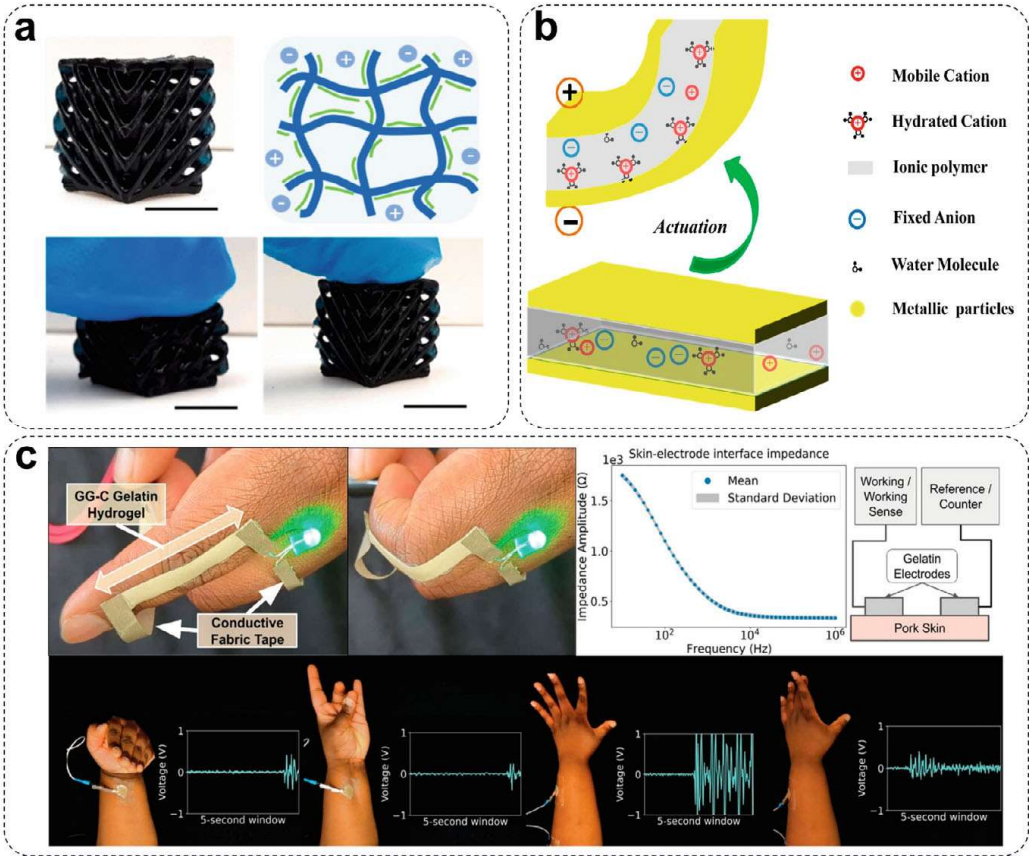


Fig. 3.5 **a** A 3D-printed conductive hydrogel (PGA coated with PEDOT:PSS) employed as a self-powered sensor. Reproduced from [45] with permission from Elsevier. **b** Schematic representation of the deformation mechanism of an IMPC under voltage. Reproduced from [55] with permission from Elsevier. **C** Hydrogel-based wearable electronics, recording signals from human muscle activation and relaxation. Reproduced from [48] under the CC BY-NC 4.0 license

in dynamic in vivo environments have also become feasible by introducing carbon nanotube (CNT) nanofillers into a semicrystalline polyvinyl alcohol (PVA) hydrogel matrix. These hydrogels demonstrated high mechanical and electrical performance, including fatigue resistance (up to 20,000 cycles at 20% strain), good stretchability (~64.5%), and low electrochemical impedance [47]. Incorporating this composite into bioelectronic devices enabled successful electromyographic (EMG) signal recordings from the spinal cord and tibialis anterior muscles of mice during optogenetic stimulation, highlighting the device's high potential for long-term neurophysiological monitoring. Furthermore, biocompatible electronics can be realized through the precise design of hydrogel composites. A biocompatible hydrogel composite composed of a percolating network of silver microflakes suspended in a natural gelatin-based matrix has been introduced [48] (see Fig. 3.5c). This composite is fabricated using inexpensive, food-safe ingredients and can be produced with consumer-grade equipment. The resulting material system is mechanically soft, stretchable up to 470%, and highly conductive (up to 3.1 $\times$ 103 S/cm).

3.2.1.3 Ionic Polymer-Metal Composites

Ionic polymer-metal composites (IPMCs) are a unique class of ion-driven electroactive polymers, typically composed of a polymeric membrane, such as Nafion or Flemion, coated with thin layers of noble metal electrodes like platinum or gold on both sides. These materials are celebrated for their lightweight structure, flexibility, and impressive deformation capabilities. Moreover, IPMCs are widely recognized as biocompatible actuators, making them suitable for various biomedical applications. The most widely accepted theory regarding the bending deformation mechanism of IPMCs involves the migration of hydrated cations. According to this theory, when a voltage is applied, hydrated cations migrate toward the cathode, aggregating and causing the polymer network near the cathode to expand. This localized expansion translates to macroscopic bending of the device, enabling its actuation (see Fig. 3.5b).

The key challenge in developing IPMCs is to ensure strong adhesion between the electrodes and the ionomer while optimizing the IPMC's response. This can be achieved through various manufacturing technologies and material choices. In most cases, a combination of technologies is used for both the electrodes and the ionomer membrane to maximize performance. A class of these fabrication technologies is electroless plating, which is commonly employed due to its high efficiency. Various methods can be used in electroless plating, including the impregnation–reduction method, the co-reduction method, and others. In the impregnation–reduction method, the polymer membrane is first immersed in a metal salt solution, allowing metal cations to infiltrate and replace the existing ions within the membrane. These metal cations are then reduced to metal particles using a reducing agent, gradually accumulating into a conductive electrode layer with enhanced adhesion and durability. The core philosophy of the co-reduction, the second electroless plating method, is to make the process more cost-effective by combining non-precious metals with precious metals during IPMC fabrication. In one study, this method

was used to deposit layers of platinum, copper, and gold as the electrode layer of an IPMC [49]. The results of 250,000-cycle life tests demonstrated good stability in the resulting actuators. This method also allows control over the interface area by adjusting the film's concentration ratio of noble and non-noble metal ions. Additionally, a novel isopropanol-assisted electroless plating technique has been introduced to create compact, nanodispersed, and homogeneously distributed metal electrodes [50]. This technique has been shown to enhance actuation performance, yielding a large bending peak-to-peak amplitude of 5.9 mm and a high tip movement speed of 224.2 mm/s (596.2°/s) at 19 Hz AC voltage. The use of isopropanol during fabrication promotes membrane swelling and enlarges ionic clusters, facilitating hydroxyl anion diffusion and simplifying metal electrode formation.

Besides electroless plating, mechanical plating technologies are also frequently used. The main advantage of these methods is their cost-effectiveness. When combined with other preparation methods, they enable the development of a cheap and fast fabrication process for IPMCs. Hot pressing is one of the mechanical plating techniques used for developing ionomer membranes. In this method, several thin membrane layers are fused together to achieve a practical polymer membrane thickness. Novel actuators composed of carbon nanotubes and chitosan membranes have been developed using hot pressing [51]. Compared to traditional IPMC actuators, these devices exhibit significant improvements in multiple aspects, including a fast response time of 19 ms across a broad frequency range and a high power density of 244 W/kg. The direct assembly method is another mechanical plating technique in which dry membranes are plated directly. However, this method cannot ensure firm adhesion between the layers. Beyond electroless and mechanical plating, new insights into the 3D printing of the ionomer layer are also being explored [52]. The promising results highlight a new direction for studying IPMCs. For a deeper understanding of IPMC fabrication methods and their variations, readers are encouraged to refer to [53]. In addition to experimental investigations, fundamental theories on IPMCs are available in the literature. For example, Panteghini and Bardella [54] modeled the electro-chemo-mechanical behavior of IPMCs based on a finite-deformation theory. Their work used a generalized finite element method (GFEM) to analyze the boundary layers near the electrodes, providing valuable insights into the precise deformation behavior of IPMCs. Given this valuable study, more versatile fundamental theories are needed for IPMCs.

3.2.2 Thermo-Responsive Polymers

Thermo-responsive polymers are common smart materials sensitive to temperature variations. Their response is typically reversible and can significantly affect properties such as solubility, shape, or mechanical behavior. This section focuses on temperature-responsive SMPs and hydrogels.

3.2.2.1 Thermo-Responsive Shape Memory Polymers

A major class of smart materials is shape memory materials, which can memorize a temporary shape and recover their original form when exposed to external stimuli [56]. Among these, SMPs—made from polymer-based materials—stand out due to their unique advantages. Compared to other shape memory materials, such as shape memory alloys, SMPs offer cost-effectiveness, low density, and high elastic deformability. Their properties can also be customized, for example, by achieving biocompatibility through structural modifications. These features make SMPs highly valuable for applications in biomedical devices, sensors, smart textiles, and, most notably, actuators. The most common external stimulus for SMPs is heat, which can be applied either directly or indirectly. Heat-sensitive SMPs are classified as thermo-responsive SMPs. Figure 3.6a illustrates the shape recovery process in a simple thermo-responsive SMP device. In addition to thermo-responsive SMPs, other types of SMPs utilize different triggering mechanisms—such as electric fields or magnetic fields—to heat the material and activate the shape memory effect. This section focuses primarily on thermo-responsive SMPs, while the final paragraph discusses electroactive SMPs (activated by an electric field). Magnetic SMPs will be covered in the following section.

Regarding conventional thermo-responsive SMPs, extensive research has been conducted on various composites and blends. The shape memory effect in PLA-TPU blends with a 30:70 composition ratio was investigated [57]. The effects of different parameters in 3D-printed fused deposition modeling (FDM) parts—such as shell thickness, infill density, and nozzle temperature—on the thermomechanical properties of the specimens were analyzed. Among these parameters, infill density had the most significant impact on shape memory performance. The PLA-TPU blend exhibited improved ductility and reduced stress relaxation compared to pure PLA, achieving shape fixity between 58 and 100% and shape recovery ranging from 53 to 91%. Another composite developed was 3D-printed PCL/TPU SMPs in bilayer and encapsulated structures [58]. In this system, PCL acted as the switching phase, while TPU served as the network point, with the melting temperature of PCL functioning as the transition temperature. The encapsulation strategy was introduced to address bonding and interface challenges between printed layers. The resulting composites demonstrated enhanced shape fixity and reduced stress relaxation. Moreover, the incorporation of magnetic nanoparticles, such as Fe_3O_4, into PETG composites not only conferred magnetic responsiveness but also improved the stiffness, recovery efficiency, and recovery speed of the material when exposed to both thermal and magnetic stimuli [59].

Beyond material innovations, 4D printing is a cutting-edge technology that enables the fabrication of SMPs with complex shapes and configurations. First introduced by Skylar Tibbits [60], 4D printing refers to the time-dependent response of 3D-printed objects to external stimuli. One of its most notable features for functional SMPs is self-assembly—the transformation of 2D patterns on flat materials into 3D structures. Origami- and kirigami-inspired SMPs have been extensively studied for their self-assembly capabilities. Xin et al. [61] investigated the shape

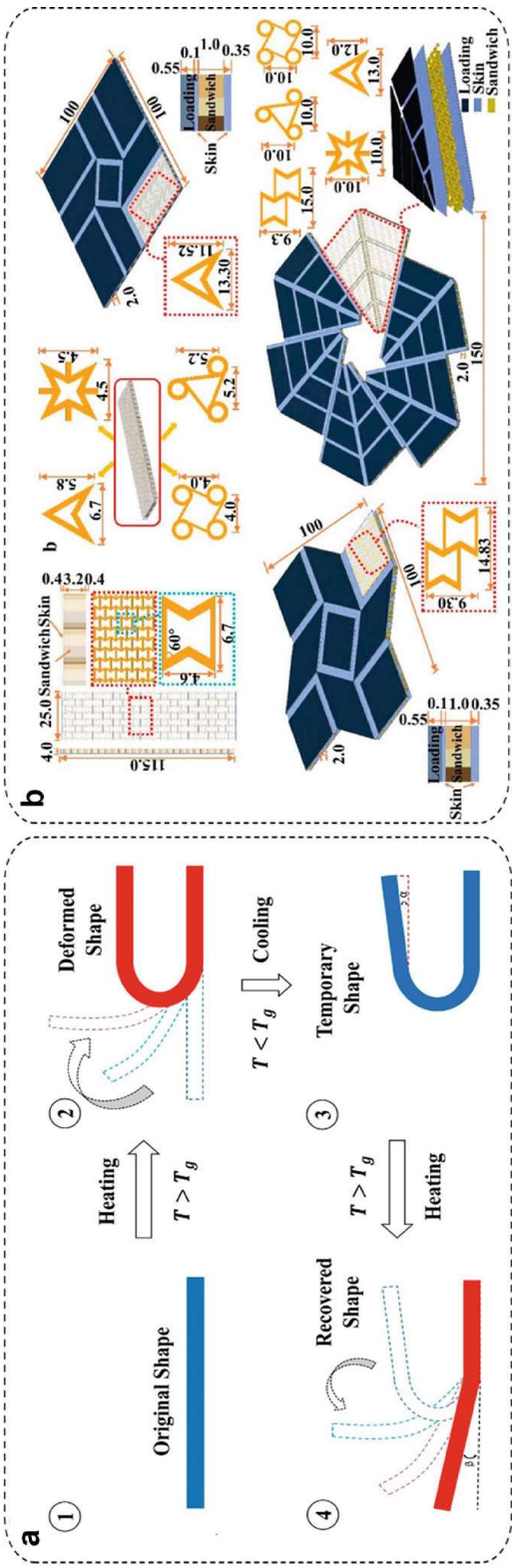


Fig. 3.6 **a** Shape recovery mechanism in dual-shape SMPs. T_g refers to the glass transition temperature. Reproduced from [68] under the CC BY-NC 4.0 license. **b** Self-deploying origami 4D-printed honeycomb sandwich structure with significant shape transformation. Reproduced from [61] with permission

memory effect in origami-inspired self-assembling SMPs with honeycomb lattice structures (see Fig. 3.6b). These active sandwich structures exhibited excellent shape memory performance, deploying within approximately one minute, which underscores their potential for antenna applications. Beyond origami-inspired 4D printing, thick-walled kirigami-based honeycomb structures with optimized geometric parameters were also fabricated, demonstrating significant volume change during shape transformation [62]. To enhance the functionality of kirigami-based devices, self-healing polycaprolactone (PCL)-based SMPs with reprocessability and thermoadaptation capabilities were developed [63], offering a versatile material for smart biomedical devices such as pyramidal scaffolds and vascular stents. In addition to self-assembly, SMPs based on TPMS have been introduced, allowing for the fabrication of highly complex architectures. Sadeghi et al. [64] investigated the shape memory properties of eight widely used TPMS meta-structures. Among them, the Schwarz-P structure demonstrated the best performance in terms of shape recovery, force recovery, and shape fixity. An optimal volume fraction of 40% was identified, providing the highest heat transfer rate and overall performance.

Apart from conventional SMPs with direct heating mechanisms, electroactive SMPs represent another class of SMPs that utilize electric fields for heating. These materials can be developed by incorporating conductive fillers, such as single-walled carbon nanotubes (SWCNTs) [65] and multiwalled carbon nanotubes (MWCNTs) [66], into traditional thermo-responsive polymers. The inclusion of SWCNTs enhances electrical conductivity and Joule heating performance; however, it may impair shape recovery. In contrast, MWCNTs offer greater thermal stability and increased stiffness, though they reduce the shape recovery efficiency of the composite. Beyond conventional fillers, novel structural composites incorporating fractal circuit patterns have also been introduced for electroactive SMPs [67]. These designs enable precise, controlled shape transformations via resistive heating. Higher-order fractal circuits promote faster and more uniform heating, significantly improving force recovery during shape memory cycles. Additionally, these composites exhibit excellent shape recovery under bending and enable innovative out-of-plane transformations through stretching.

3.2.2.2 Thermo-Responsive Hydrogels

Thermo-responsive hydrogels are polymer networks that undergo significant and reversible volume changes in response to temperature variations. Based on phase separation phenomena, these hydrogels exhibit either negative (inverse) or positive thermo-sensitivity. Hydrogels in the first group become insoluble in water when the surrounding temperature exceeds their lower critical solution temperature (LCST), while those in the second group show increased solubility above their upper critical solution temperature (UCST). The operational mechanism of thermo-responsive hydrogels is straightforward, allowing easy control and accurate prediction of their response. Their transition temperature can be tailored by modifying the polymer composition, making them suitable for a wide range of applications.

Notably, thermo-responsive hydrogels with an LCST near body temperature (e.g., poly(N-isopropylacrylamide) or PNIPAM) are extensively developed for use in drug delivery systems, tissue engineering scaffolds, and soft robotics (see Fig. 3.7).

In the field of biomedical devices, hydrogel-based contact lenses (HCLs) have demonstrated significant practicality. HCLs offer improved wettability, oxygen permeability, adaptability, and comfort compared to rigid contact lenses. Additionally, they exhibit greater safety and enhanced ocular tissue compatibility compared to traditional contact lenses. Highly transparent and biocompatible HCLs, composed of a PVA matrix reinforced with nanocellulose, have been developed [69]. These HCLs achieve high water content (>90%) and exceptional optical properties without compromising mechanical performance. In the context of tissue engineering, 3D chitosan bioscaffolds have been proposed for chronic wound healing applications [70]. These bioscaffolds demonstrated excellent biocompatibility and cytocompatibility toward two skin-associated human cell lines, promoting faster tissue regeneration and offering cost-effective fabrication. Similarly, biocompatible cellulose nanofibril (CNF) hydrogels were developed through 3D printing techniques [71]. The resulting specimens exhibited a tunable mechanical stiffness range (3–8 kPa), making them promising candidates for biomedical applications. Hydrogel composites can be engineered to exhibit enhanced mechanical properties, which is crucial for soft actuators and grippers. In this regard, tough Al-alginate/ PNIPAM hydrogels were synthesized using interpenetrating polymer network IPN structures comprising hybrid physically cross-linked alginate and chemically cross-linked PNIPAM [72]. These hydrogel composites achieved a remarkable compressive stress of 6.3 ± 0.3 MPa and a uniaxial stretch up to 9.95. Both the mechanical properties and the LCST of the composite could be tuned by modifying its composition.

Apart from the experimental advancements in hydrogel composites, analytical solutions for the response of both pristine and composite hydrogels in practical problems have also been developed. Herein, we briefly introduce some notable analytical studies on hydrogels. Theoretical studies of responsive hydrogels have mostly been developed based on principles of nonlinear solid mechanics, providing a robust framework for investigating their extreme mechanical responses. The typical approach to capturing such behavior involves satisfying the equilibrium equations while assigning an appropriate strain energy function to the material medium. The equilibrium state of the material can be obtained by finding the stationary points of this function. Following this theoretical approach, early studies successfully modeled the coupled fluid permeation and large deformation behavior of neutral hydrogels in several benchmark problems [75–77]. Moreover, determining the equilibrium state of the material via strain energy function derivation can be employed for modeling hydrogel composites. This can be achieved by adding regulatory terms to the defined strain energy function, representing the strain energy contribution of the reinforcement phase added to pristine hydrogels. Studies following this approach have addressed the swelling behavior of hydrogel composites with rigid cores and hydrogel bilayers [78]. A mathematical model for reinforced hydrogels with wood-based microfibers has also been developed recently [79].

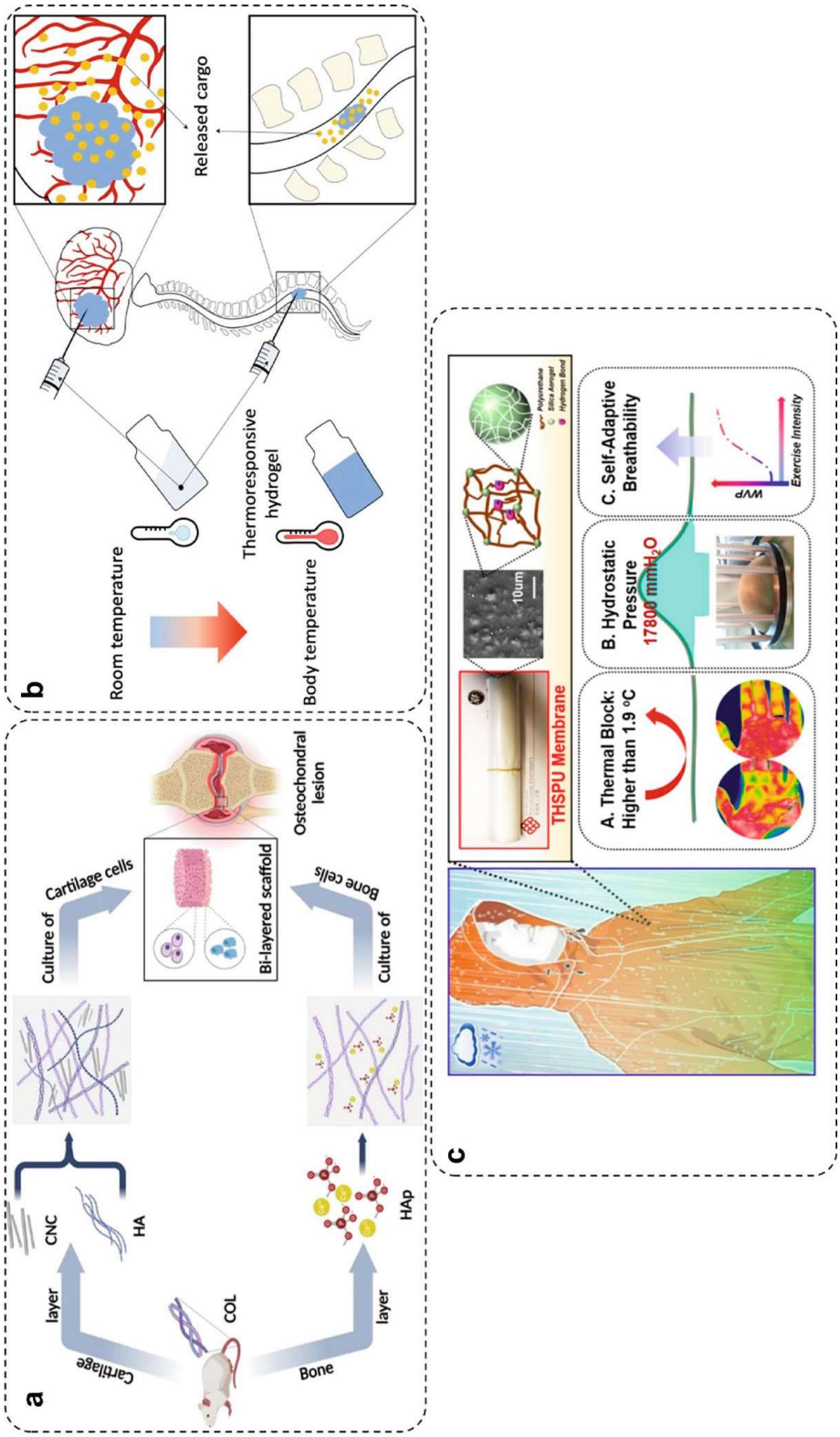


Fig. 3.7 **a** Schematics of the fabrication process for a collagen-based scaffold mimicking osteochondral tissue functions. Reproduced from [73] under the CC BY-NC-ND 4.0 license. **b** Injectable thermo-responsive hydrogels and their use in drug delivery systems. The composite's structure is illustrated across multiple scales. Reproduced from [74] under the CC BY-NC-ND 4.0 license. **c** Waterproof and self-adaptive breathable smart textiles based on polyurethane membranes. Reproduced from with permission [6]

Modification of the strain energy function is also a useful approach for hydrogels with stimulus responsiveness. These additional terms can serve as constraints that capture specific physical phenomena in the system. For instance, in thermo-responsive hydrogels, adding an additional term—called the *mixing energy term*—to the strain energy function captures the energy change of the hydrogel medium during mixing with the solvent. This term correlates temperature and volume fraction variations, thereby introducing thermo-responsiveness into the material. To date, both analytical and numerical solutions for temperature-sensitive hydrogels with composite structures have been presented in numerous studies [80, 81]. This scenario—adding modifying terms to the strain energy function—can similarly be applied to capture large deformations in magneto-responsive hydrogels [82]. For ionic hydrogels, however, modification of the strain energy function does not seem to be sufficient. Instead, a coupled multiphysics problem must be defined, in which two conservation laws are satisfied simultaneously to ensure equilibrium in both the mechanical and chemical aspects of the system. Such a theoretical framework has been previously developed in [83]. More recent studies in this field focus on numerical solutions for static and dynamic regimes in pH-responsive hydrogels with composite structures, advancing the pre-design of soft actuators for acidic or basic environments [84, 85]. Investigations have also been conducted on the equilibrium swelling behavior of cellulose-reinforced, salt-sensitive hydrogels [86]. The emerging direction of current research is to explore the mesoscale properties of responsive hydrogels and to propose modifications to diffusion theories for such materials [87, 88]. These studies facilitate the design of next-generation responsive hydrogels with more controlled deformations, which are highly beneficial for drug delivery system design.

3.2.3 Magneto-Responsive Polymers

Magnetically responsive materials are primarily composites composed of soft polymers embedded with magnetic particles. When exposed to a magnetic field, the particles become magnetized and align with the field's direction. Since magnetic fields can penetrate various media, magnetically responsive actuators are ideal for operation in confined spaces and are widely used in untethered microrobots. This section briefly covers magnetic shape memory polymers, magnetic hydrogels, and magnetorheological elastomers. Additionally, a concise overview of theoretical studies on smart hydrogels is provided.

3.2.3.1 Magneto-Responsive Shape Memory Polymers

It has been previously stated that, except for traditional SMPs that rely on direct heating mechanisms, novel SMPs with indirect heating mechanisms have been developed. A significant subclass of these materials is magnetic shape memory polymers (MSMPs). MSMPs have attracted considerable interest in the biomedical field due

to their ability to be remotely controlled without requiring direct heat, enhancing their safety and applicability. When an MSMP is exposed to an external alternating magnetic field, the embedded magnetic particles generate heat through induction heating. Once the activation temperature is reached, the shape memory effect is triggered.

Various magnetic particles have been incorporated into MSMPs, with Fe_3O_4 nanoparticles being among the most commonly used. Several SMP formulations embedded with Fe_3O_4 nanoparticles have been proposed. For instance, Yu et al. [89] developed an MSMP based on poly(ε-caprolactone) (c-PCL) incorporating Fe_3O_4 nanoparticles (5–25 wt%). The crosslinking schemes employed in their study enhanced the polymer's shape recovery by up to 70%. Their MSMP also exhibited multifunctional activation through both hot water and an alternating magnetic field. However, a faster shape recovery time was observed in hot water due to higher thermal transfer efficiency and reduced heat loss. Also, 4D-printed MSMPs composed of polymethyl methacrylate (PMMA) and thermoplastic polyurethane (TPU), embedded with Fe_3O_4 nanoparticles, have been investigated [90]. In these composites, PMMA provides structural rigidity, TPU enhances flexibility, and the Fe_3O_4 nanoparticles enable magnetic responsiveness, allowing controlled shape transformations (see Fig. 3.8a). In another study, Yousefi et al. [68] developed 4D-printed PLA-PBAT composites reinforced with Fe_3O_4 nanoparticles at varying concentrations (10, 15, and 20%). These composites demonstrated improved mechanical properties and shape memory behavior without compromising printability, making them promising candidates for high-performance, sustainable structures with remote activation capabilities.

In addition to Fe_3O_4 nanoparticles, MSMPs incorporating other magnetic fillers, such as nickel-zinc ferrite [91] and carbonyl iron particles (CIPs) [92], have also been studied. Research has shown that SMP specimens containing up to 10 vol% of nickel-zinc ferrite particles can achieve uniform and rapid heating without compromising shape recovery performance. Beyond SMP composites with embedded magnetic fillers, a novel approach has been proposed—combining magnetic and conductive PLA composites with FDM technology for 4D printing [93]. The resulting composite can be controlled via magnetic fields, exhibiting significant potential for high-temperature applications. It also demonstrated the ability to achieve multiple shape configurations and repeatedly switch between temporary and permanent shapes, while minimizing material waste. In addition to material innovations, MSMPs with lattice structures represent another area of research, offering advanced structural composites. 3D-printed PLA-MXene composites with skeletal and sheet-based TPMS gyroid structures have been explored for their shape memory properties. Results indicate that adding up to 0.5 wt% MXene to PLA significantly improves both stiffness and energy absorption capacity [94].

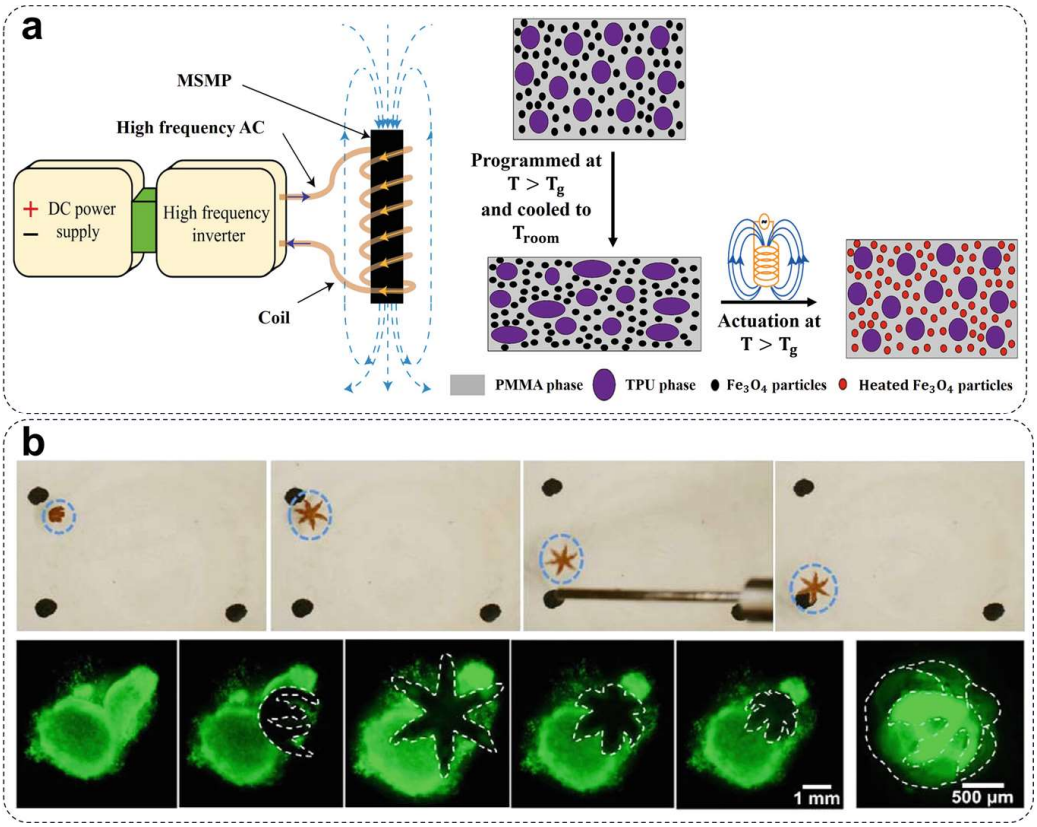


Fig. 3.8 **a** Induction heating setup used for heating up PMMA/TPU/Fe_3O_4 Shape memory polymer composites. Adapted from [90] under the CC BY-NC 4.0 license. Shape changing mechanism is shown. **b** Applications of soft thermo-responsive self-folding microgrippers for precise magnetic guidance, cell capture, and excision from a live fibroblast clump. Reproduced from [99] with permission. Copyright © 2015 American Chemical Society

3.2.3.2 Magnetorheological Elastomers

Magnetorheological elastomers (MREs) are a type of magneto-responsive polymers that consist of an elastomeric or rubber-based matrix. The dynamic properties of MREs can be modified and tailored by applying an external magnetic field. Unlike other magneto-responsive polymers, the magnetorheological effect in MREs provides a rapid and reversible response, making them ideal for applications such as tunable vibration absorbers and materials with magnetically controlled stiffness. So far, the magnetorheological properties of MRE composites reinforced with carbonyl iron particles at various concentrations have been investigated both numerically and experimentally [95, 96]. Furthermore, as a novel approach offering great design flexibility, 3D printing has been utilized to create self-sensing MRE composites [97]. A thermoplastic elastomer (SEBS-based TPS) was used as the matrix. At the same time, hybrid fillers (carbon black and CIP) were incorporated to combine magnetostrictive, magnetoresistive, and piezoresistive properties in a single material. However, 3D printing strategies for MRE composites are still under development. Studies

have shown that different printing strategies can yield different magnetorheological effects. For instance, previous findings indicate that circular printing paths enhance the MR effect compared to linear printing paths [98].

3.2.3.3 Magneto-Responsive Hydrogels

By incorporating magnetic fillers such as Fe_3O_4, magneto-responsive hydrogels can be readily fabricated. These magnetic hydrogels have practical applications in microgrippers, medical devices, microfluidics, and drug delivery systems. For instance, star-shaped functional microgrippers capable of untethered grasping and clamping of microscopic objects—such as cancer cells and tumors—were demonstrated by developing pNIPAM-AAc hydrogels reinforced with a non-swellable, stiff segmented polymer to enhance gripping performance [99]. Embedding Fe_2O_3 nanoparticles into the hydrogel enabled remote control of the microgrippers via external magnetic fields (see Fig. 3.8b). Strong, tough, and stretchable magnetic hydrogels were also achieved by introducing hybrid physical and chemical crosslinks into Fe_3O_4-alginate/polyacrylamide hydrogels [100]. One notable example of these hydrogel nanocomposites is their application in magnetically guided catheters, demonstrating their potential in biomedical engineering.

Self-adaptive magnetic photonic nanochain cilia arrays have also been developed by combining magnetic nanoparticles with pH-responsive hydrogels [101]. These devices represent significant advancements in intelligent microfluidic systems, enabling real-time pH sensing and self-adaptive fluid pumping. They offer visual feedback in response to pH variations for microenvironment detection and allow for flow velocity adjustments under a rotating magnetic field. Also, precise 3D navigation has been achieved by creating soft helical hydrogel motors composed of PVA and magnetic nanoparticles [102]. These biocompatible motors can be easily fabricated and guided using a low-strength rotating magnetic field, facilitating targeted drug delivery. This innovative motor introduces a novel approach to directing immune cell migration and organizing immune responses, with the potential to revolutionize the biomedical industry and disease treatment strategies.

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References

1. A. Fattah-alhosseini, R. Chaharmahali, S. Alizad, M. Kaseem, B. Dikici, A review of smart polymeric materials: recent developments and prospects for medicine applications. Hybrid Adv. **5**, 100178 (2024). https://doi.org/10.1016/J.HYBADV.2024.100178

2. S.L. Duraikkannu, R. Castro-Muñoz, A. Figoli, A review on phase-inversion technique-based polymer microsphere fabrication. Colloid Interface Sci. Commun. **40**, 100329 (2021). https://doi.org/10.1016/j.colcom.2020.100329
3. X. Zhao, X. Chen, H. Yuk, S. Lin, X. Liu, G. Parada, Soft materials by design: unconventional polymer networks give extreme properties. Chem. Rev. **121**(8), 4309–4372 (2021). https://doi.org/10.1021/ACS.CHEMREV.0C01088
4. N. El-Atab et al., Soft Actuators for soft robotic applications: a review. Adv. Intell. Syst. **2**(10), 2000128 (2020). https://doi.org/10.1002/aisy.202000128
5. F. Arab Hassani et al., Smart materials for smart healthcare– moving from sensors and actuators to self-sustained nanoenergy nanosystems. Smart Mater. Med. **1**, 92–124 (2020). https://doi.org/10.1016/j.smaim.2020.07.005
6. S. Shi, Y. Han, J. Hu, Robust waterproof and self-adaptive breathable membrane with heat retention property for intelligent protective cloth. Prog. Org. Coat. **137**, 105303 (2019). https://doi.org/10.1016/J.PORGCOAT.2019.105303
7. X.A. Zhang et al., Dynamic gating of infrared radiation in a textile. Science **363**(6427), 619–623 (2019). https://doi.org/10.1126/SCIENCE.AAU1217/SUPPL_FILE/AAU1217S1.MP4
8. M.T. Flavin et al., Bioelectronics for targeted pain management. Nat. Rev. Electric. Eng. **2**(6), 407–424 (2025). https://doi.org/10.1038/S44287-025-00177-3
9. M.A. Moreno-Mateos, M.L. Lopez-Donaire, M. Hossain, D. Garcia-Gonzalez, Effects of soft and hard magnetic particles on the mechanical performance of ultra-soft magnetorheological elastomers. Smart Mater. Struct. **31**(6), 065018 (2022). https://doi.org/10.1088/1361-665X/AC6BD3
10. S.R. Shin et al., Carbon-nanotube-embedded hydrogel sheets for engineering cardiac constructs and bioactuators. ACS Nano **7**(3), 2369–2380 (2013). https://doi.org/10.1021/NN305559J/SUPPL_FILE/NN305559J_SI_006.MPG
11. M. Duduta, E. Hajiesmaili, H. Zhao, R.J. Wood, D.R. Clarke, Realizing the potential of dielectric elastomer artificial muscles. Proc. Natl. Acad. Sci. **116**(7), 2476–2481 (2019). https://doi.org/10.1073/pnas.1815053116
12. D. Pyo, S. Ryu, K.-U. Kyung, S. Yun, D.-S. Kwon, High-pressure endurable flexible tactile actuator based on microstructured dielectric elastomer. Appl. Phys. Lett. **112**(6), 061902 (2018). https://doi.org/10.1063/1.5016385
13. W. Feng et al., A large-strain and ultrahigh energy density dielectric elastomer for fast moving soft robot. Nat. Commun. **15**(1), 4222 (2024). https://doi.org/10.1038/s41467-024-48243-y
14. M.N. Ismail, A.I. Khalaf, Styrene–butadiene rubber/graphite powder composites: Rheometrical, physicomechanical, and morphological properties. J. Appl. Polym. Sci. **120**(1), 298–304 (2011). https://doi.org/10.1002/app.33101
15. H. Liu, Y. Shen, Y. Song, C.-W. Nan, Y. Lin, X. Yang, Carbon nanotube array/polymer core/shell structured composites with high dielectric permittivity, low dielectric loss, and large energy density. Adv. Mater. **23**(43), 5104–5108 (2011). https://doi.org/10.1002/adma.201102079
16. W. Hu, S.N. Zhang, X. Niu, C. Liu, Q. Pei, An aluminum nanoparticle–acrylate copolymer nanocomposite as a dielectric elastomer with a high dielectric constant. J Mater. Chem. C Mater. **2**(9), 1658–1666 (2014). https://doi.org/10.1039/C3TC31929F
17. M. Razzaghi Kashani, S. Javadi, N. Gharavi, Dielectric properties of silicone rubber-titanium dioxide composites prepared by dielectrophoretic assembly of filler particles. Smart Mater Struct. **19**(3), 035019 (2010). https://doi.org/10.1088/0964-1726/19/3/035019
18. L.J. Romasanta et al., Towards materials with enhanced electro-mechanical response: CaCu3Ti4O12–polydimethylsiloxane composites. J. Mater. Chem. **22**(47), 24705–24712 (2012). https://doi.org/10.1039/C2JM34674E
19. X. Niu, H. Stoyanov, W. Hu, R. Leo, P. Brochu, Q. Pei, Synthesizing a new dielectric elastomer exhibiting large actuation strain and suppressed electromechanical instability without prestretching. J. Polym. Sci. B Polym. Phys. **51**(3), 197–206 (2013). https://doi.org/10.1002/polb.23197

20. M.A. Safaei, M. Baghani, M. Baniassadi, M. Bodaghi, In silico actuation performance investigation of dielectric elastomers with TPMS geometries. Euro. J. Mech. A/Solids 105540 (2024). https://doi.org/10.1016/j.euromechsol.2024.105540
21. A. Chortos, J. Mao, J. Mueller, E. Hajiesmaili, J.A. Lewis, D.R. Clarke, Printing reconfigurable bundles of dielectric elastomer fibers. Adv. Funct. Mater. **31**(22), 2010643 (2021). https://doi.org/10.1002/adfm.202010643
22. J. Ni et al., Strong fatigue-resistant nanofibrous hydrogels inspired by lobster underbelly. Matter **4**(6), 1919–1934 (2021). https://doi.org/10.1016/j.matt.2021.03.023
23. D. McCoul, S. Rosset, S. Schlatter, H. Shea, Inkjet 3D printing of UV and thermal cure silicone elastomers for dielectric elastomer actuators. Smart Mater. Struct. **26**(12), 125022 (2017). https://doi.org/10.1088/1361-665X/aa9695
24. A. Chortos, E. Hajiesmaili, J. Morales, D.R. Clarke, J.A. Lewis, 3D printing of interdigitated dielectric elastomer actuators. Adv. Funct. Mater. **30**(1), 1907375 (2020). https://doi.org/10.1002/adfm.201907375
25. E. Hajiesmaili, D.R. Clarke, Reconfigurable shape-morphing dielectric elastomers using spatially varying electric fields. Nat. Commun. **10**(1), 183 (2019). https://doi.org/10.1038/s41467-018-08094-w
26. E. Hajiesmaili, E. Khare, A. Chortos, J. Lewis, D.R. Clarke, Voltage-controlled morphing of dielectric elastomer circular sheets into conical surfaces. Extreme Mech. Lett. **30**, 100504 (2019). https://doi.org/10.1016/j.eml.2019.100504
27. E. Hajiesmaili, N.M. Larson, J.A. Lewis, D.R. Clarke, Programmed shape-morphing into complex target shapes using architected dielectric elastomer actuators. Sci. Adv. **8**(28), eabn9198 (2024). https://doi.org/10.1126/sciadv.abn9198
28. Z. Suo, Theory of dielectric elastomers. Acta Mech. Solida Sin. **23**(6), 549–578 (2010). https://doi.org/10.1016/S0894-9166(11)60004-9
29. E. Hajiesmaili, D.R. Clarke, Dielectric elastomer actuators. J. Appl. Phys. **129**(15), 151102 (2021). https://doi.org/10.1063/5.0043959
30. J. Zhou, W. Hong, X. Zhao, Z. Zhang, Z. Suo, Propagation of instability in dielectric elastomers. Int. J. Solids Struct. **45**(13), 3739–3750 (2008). https://doi.org/10.1016/j.ijsolstr.2007.09.031
31. M. Bahreman, N. Arora, H. Darijani, S. Rudykh, Structural and material electro-mechanical instabilities in microstructured dielectric elastomer plates. Eur. J. Mech. A. Solids **94**, 104534 (2022). https://doi.org/10.1016/j.euromechsol.2022.104534
32. M. Majidi, M. Asgari, Nonlinear bending–twisting coupling in electromechanical finite deformation of fiber-reinforced tubular dielectric elastomer for soft actuators. Int. J. Non Linear Mech. **156** (2023). https://doi.org/10.1016/j.ijnonlinmec.2023.104480
33. M.A. Alioglu et al., A versatile photocrosslinkable silicone composite for 3d printing applications. Adv. Mater. Technol. **9**(3), 2301858 (2024). https://doi.org/10.1002/admt.202301858
34. V. Karamzadeh et al., High-resolution additive manufacturing of a biodegradable elastomer with a low-cost LCD 3D printer. Adv. Healthc. Mater. **13**(9), 2303708 (2024). https://doi.org/10.1002/adhm.202303708
35. E.M. Ahmed, Hydrogel: preparation, characterization, and applications: a review. J. Adv. Res. **6**(2), 105–121 (2015). https://doi.org/10.1016/j.jare.2013.07.006
36. L. Wang, M. Zhou, T. Xu, X. Zhang, Multifunctional hydrogel as wound dressing for intelligent wound monitoring. Chem. Eng. J. **433**, 134625 (2022). https://doi.org/10.1016/j.cej.2022.134625
37. D.J. Beebe et al., Functional hydrogel structures for autonomous flow control inside microfluidic channels. Nature **404**(6778), 588–590 (2000). https://doi.org/10.1038/35007047
38. J. Zhao, X. Zhao, B. Guo, P.X. Ma, Multifunctional interpenetrating polymer network hydrogels based on methacrylated alginate for the delivery of small molecule drugs and sustained release of protein. Biomacromol **15**(9), 3246–3252 (2014). https://doi.org/10.1021/bm5006257

39. G. Zhang, J. Steck, J. Kim, C. H. Ahn, Z. Suo, Hydrogels of arrested phase separation simultaneously achieve high strength and low hysteresis. Sci. Adv. **9**(26), eadh7742 (2024). https://doi.org/10.1126/sciadv.adh7742
40. H. Zhuo et al., Bamboo-inspired ultra-strong nanofiber-reinforced composite hydrogels. Nat. Commun. **16**(1), 980 (2025). https://doi.org/10.1038/s41467-025-56340-9
41. R. Liu et al., Preparation and characterization of multilayer ph-responsive hydrogel loaded ganoderma lucidum peptides. Foods **12**(7) (2023). https://doi.org/10.3390/foods12071481
42. T.M. Valentin, E.M. DuBois, C.E. Machnicki, D. Bhaskar, F.R. Cui, I.Y. Wong, 3D printed self-adhesive PEGDA–PAA hydrogels as modular components for soft actuators and microfluidics. Polym. Chem. **10**(16), 2015–2028 (2019). https://doi.org/10.1039/C9PY00211A
43. K.M. Lichade, S. Shiravi, J.D. Finan, Y. Pan, Direct printing of conductive hydrogels using two-photon polymerization. Addit. Manuf. **84**, 104123 (2024). https://doi.org/10.1016/j.addma.2024.104123
44. L. Zhang et al., 3D direct printing of mechanical and biocompatible hydrogel meta-structures. Bioact. Mater. **10**, 48–55 (2022). https://doi.org/10.1016/j.bioactmat.2021.08.015
45. R. Yang et al., 3D-printed conducting polymer hydrogel-based DC generator for self-powered electromechanical sensing. Nano Energy **117**, 108857 (2023). https://doi.org/10.1016/j.nanoen.2023.108857
46. Y. Ohm, C. Pan, M.J. Ford, X. Huang, J. Liao, C. Majidi, An electrically conductive silver–polyacrylamide–alginate hydrogel composite for soft electronics. Nat. Electron **4**(3), 185–192 (2021). https://doi.org/10.1038/s41928-021-00545-5
47. S. Huang et al., Anisotropic hydrogel microelectrodes for intraspinal neural recordings in vivo. Nat. Commun. **16**(1), 1127 (2025). https://doi.org/10.1038/s41467-025-56450-4
48. K. Chin, M. Vinciguerra, D.K. Patel, P. Roberts Olcay, E.S.L. Vasquez, C. Majidi, accessible soft electronics with silver-gelatin conductive hydrogel composite. Adv. Mater. Technol. 2401193 (2024). https://doi.org/10.1002/admt.202401193
49. M.D. Bennett, D.J. Leo, Manufacture and characterization of ionic polymer transducers employing non-precious metal electrodes. Smart Mater. Struct. **12**(3), 424 (2003). https://doi.org/10.1088/0964-1726/12/3/314
50. S. Ma, Y. Zhang, Y. Liang, L. Ren, W. Tian, L. Ren, High-performance ionic-polymer–metal composite: toward large-deformation fast-response artificial muscles. Adv. Funct. Mater. **30**(7), 1908508 (2020). https://doi.org/10.1002/adfm.201908508
51. J. Li et al., Superfast-response and ultrahigh-power-density electromechanical actuators based on hierarchal carbon nanotube electrodes and chitosan. Nano Lett. **11**(11), 4636–4641 (2011). https://doi.org/10.1021/nl202132m
52. J.D. Carrico, N.W. Traeden, M. Aureli, K.K. Leang, Fused filament 3D printing of ionic polymer-metal composites (IPMCs). Smart Mater. Struct. **24**(12), 125021 (2015). https://doi.org/10.1088/0964-1726/24/12/125021
53. C. He et al., Preparation and modification technology analysis of ionic polymer-metal composites (IPMCs). Int. J. Mol. Sci. **23**(7) (2022). https://doi.org/10.3390/ijms23073522
54. A. Panteghini, L. Bardella, Electrochemo-poromechanics of ionic polymer metal composites: towards the accurate finite element modelling of actuation and sensing. J. Elast. **153**(3), 299–358 (2023). https://doi.org/10.1007/s10659-023-09990-z
55. L. Yang, D. Zhang, X. Zhang, A. Tian, Surface profile topography of ionic polymer metal composite based on fractal theory. Surf. Interfac. **22**, 100834 (2021). https://doi.org/10.1016/j.surfin.2020.100834
56. P. Honari et al., Simultaneous 4D printing and In-Situ foaming for tailoring shape memory behaviors in polylactic acid foams. Manuf. Lett. **43**, 33–36 (2025). https://doi.org/10.1016/j.mfglet.2024.11.005
57. D. Rahmatabadi et al., Shape memory performance assessment of FDM 3D printed PLA-TPU composites by Box-Behnken response surface methodology. Int. J. Adv. Manufact. Technol. **127**(1), 935–950 (2023). https://doi.org/10.1007/s00170-023-11571-2
58. D. Rahmatabadi et al., 4D printing-encapsulated polycaprolactone-thermoplastic polyurethane with high shape memory performances. Adv. Eng. Mater. **25**(6), 2201309 (2023). https://doi.org/10.1002/adem.202201309

59. D. Rahmatabadi et al., 4D printing thermo-magneto-responsive PETG-Fe3O4 nanocomposites with enhanced shape memory effects. Appl. Mater. Today **40**, 102361 (2024). https://doi.org/10.1016/j.apmt.2024.102361
60. S. Tibbits, 4D printing: multi-material shape change. Archit. Des. **84**(1), 116–121 (2014). https://doi.org/10.1002/ad.1710
61. X. Xin, L. Liu, Y. Liu, J. Leng, Origami-inspired self-deployment 4D printed honeycomb sandwich structure with large shape transformation. Smart Mater. Struct. **29**(6), 065015 (2020). https://doi.org/10.1088/1361-665X/ab85a4
62. C. Yue, W. Zhao, F. Li, L. Liu, Y. Liu, J. Leng, Shape recovery properties and load-carrying capacity of a 4D printed thick-walled kirigami-inspired honeycomb structure. Biodes Manuf. **6**(2), 189–203 (2023). https://doi.org/10.1007/s42242-022-00230-2
63. X. Li, L. Wang, Y. Li, S. Xu, Reprocessable, self-healing, thermadapt shape memory polycaprolactone via robust ester-ester interchanges toward kirigami-tailored 4d medical devices. ACS Appl. Polym. Mater. **5**(2), 1585–1595 (2023). https://doi.org/10.1021/acsapm.2c02070
64. F. Sadeghi, M. Baniassadi, A. Shahidi, M. Baghani, TPMS metamaterial structures based on shape memory polymers: mechanical, thermal and thermomechanical assessment. J. Mater. Res. Technol. **23**, 3726–3743 (2023). https://doi.org/10.1016/j.jmrt.2023.02.014
65. Y.-C. Sun, M. Chu, M. Huang, O. Hegazi, H.E. Naguib, Hybrid electroactive shape memory polymer composites with room temperature deformability. Macromol. Mater. Eng. **304**(10), 1900196 (2019). https://doi.org/10.1002/mame.201900196
66. F.-P. Du et al., Electroactive shape memory polymer based on optimized multi-walled carbon nanotubes/polyvinyl alcohol nanocomposites. Compos. B Eng. **68**, 170–175 (2015). https://doi.org/10.1016/j.compositesb.2014.08.043
67. A. Enferadi, M. Baniassadi, M. Baghani, Fractal circuit architectures for spatially controlled heating and multi-modal shape programming of electro-active shape memory polymers. Int. J. Appl. Mech. **16**(06), 2441003 (2024). https://doi.org/10.1142/S1758825124410035
68. M.A. Yousefi, D. Rahmatabadi, M. Baniassadi, M. Bodaghi, M. Baghani, 4D printing of multifunctional and biodegradable pla-pbat-fe3o4 nanocomposites with supreme mechanical and shape memory properties. Macromol. Rapid Commun. **46**(2), 2400661 (2025). https://doi.org/10.1002/marc.202400661
69. G.K. Tummala, R. Rojas, A. Mihranyan, Poly(vinyl alcohol) hydrogels reinforced with nanocellulose for ophthalmic applications: general characteristics and optical properties. J. Phys. Chem. B **120**(51), 13094–13101 (2016). https://doi.org/10.1021/acs.jpcb.6b10650
70. C. Intini et al., 3D-printed chitosan-based scaffolds: an in vitro study of human skin cell growth and an in-vivo wound healing evaluation in experimental diabetes in rats. Carbohydr. Polym. **199**, 593–602 (2018). https://doi.org/10.1016/j.carbpol.2018.07.057
71. C. Xu et al., 3D printing of nanocellulose hydrogel scaffolds with tunable mechanical strength towards wound healing application. J. Mater. Chem. B **6**(43), 7066–7075 (2018). https://doi.org/10.1039/C8TB01757C
72. W.J. Zheng, N. An, J.H. Yang, J. Zhou, Y.M. Chen, Tough Al-alginate/Poly(N-isopropylacrylamide) hydrogel with tunable LCST for soft robotics. ACS Appl. Mater. Interfaces **7**(3), 1758–1764 (2015). https://doi.org/10.1021/am507339r
73. F. Mahmoudi, S. Takallu, M.A. Derakhshan, E. Mirzaei, A collagen-based bilayer nanocomposite scaffold for osteochondral tissue regeneration: integrating hydroxyapatite and cellulose nanocrystals. J. Mater. Sci. Mater. Med. **36**(1), 88 (2025). https://doi.org/10.1007/S10856-025-06953-9/FIGURES/8
74. E. Bellotti, A.L. Schilling, S.R. Little, P. Decuzzi, Injectable thermoresponsive hydrogels as drug delivery system for the treatment of central nervous system disorders: a review. J. Control. Release **329**, 16–35 (2021). https://doi.org/10.1016/j.jconrel.2020.11.049
75. S.A. Chester, C.V. Di Leo, L. Anand, A finite element implementation of a coupled diffusion-deformation theory for elastomeric gels. Int. J. Solids Struct. **52**, 1–18 (2015). https://doi.org/10.1016/j.ijsolstr.2014.08.015
76. S.A. Chester, L. Anand, A coupled theory of fluid permeation and large deformations for elastomeric materials. J. Mech. Phys. Solids **58**(11), 1879–1906 (2010). https://doi.org/10.1016/j.jmps.2010.07.020

77. A. Lucantonio, P. Nardinocchi, L. Teresi, Transient analysis of swelling-induced large deformations in polymer gels. J. Mech. Phys. Solids **61**(1), 205–218 (2013). https://doi.org/10.1016/j.jmps.2012.07.010
78. X. Zhao, W. Hong, Z. Suo, Inhomogeneous and anisotropic equilibrium state of a swollen hydrogel containing a hard core. Appl. Phys. Lett. **92**(5), 051904 (2008). https://doi.org/10.1063/1.2840158
79. J. Cheng, Z. Jia, T. Li, A constitutive model of microfiber reinforced anisotropic hydrogels: with applications to wood-based hydrogels. J. Mech. Phys. Solids **138**, 103893 (2020). https://doi.org/10.1016/j.jmps.2020.103893
80. H. Mazaheri, M. Baghani, R. Naghdabadi, S. Sohrabpour, Inhomogeneous swelling behavior of temperature sensitive PNIPAM hydrogels in micro-valves: analytical and numerical study. Smart Mater. Struct. **24**(4), 045004 (2015). https://doi.org/10.1088/0964-1726/24/4/045004
81. H. Mazaheri, M. Baghani, R. Naghdabadi, S. Sohrabpour, Coupling behavior of the pH/temperature sensitive hydrogels for the inhomogeneous and homogeneous swelling. Smart Mater. Struct. **25**(8), 085034 (2016). https://doi.org/10.1088/0964-1726/25/8/085034
82. R. Zhao, Y. Kim, S.A. Chester, P. Sharma, X. Zhao, Mechanics of hard-magnetic soft materials. J. Mech. Phys. Solids **124**, 244–263 (2019). https://doi.org/10.1016/J.JMPS.2018.10.008
83. S.K. De, N.R. Aluru, A chemo-electro-mechanical mathematical model for simulation of pH sensitive hydrogels. Mech. Mater. **36**(5–6), 395–410 (2004). https://doi.org/10.1016/S0167-6636(03)00067-X
84. M. Askari-Sedeh, M. Baghani, pH-sensitive hydrogel bilayers: investigation on transient swelling-induced bending through analytical and FEM approaches. Gels **9**(7) (2023). https://doi.org/10.3390/gels9070563
85. M. Askari-Sedeh, M. Baghani, Coupled chemo-mechanical swelling behavior of ph-sensitive hollow cylinder hydrogels under extension–torsion and internal pressure: analytical and 3D FEM solutions. **15**(4) (2023). https://doi.org/10.1142/S1758825123500308
86. H.R. Safavi, A. Amiri, M. Baniassadi, A. Zolfagharian, M. Baghani, An anisotropic constitutive model for fiber reinforced salt-sensitive hydrogels. Mech. Adv. Mater. Struct. **30**(23), 4814–4827 (2023). https://doi.org/10.1080/15376494.2022.2106523
87. R.J. Wagner, J. Dai, X. Su, F.J. Vernerey, A mesoscale model for the micromechanical study of gels. J. Mech. Phys. Solids **167**, 104982 (2022). https://doi.org/10.1016/J.JMPS.2022.104982
88. J. Liu, S. Lin, Strain-engineered particle diffusion in uniaxially deformed polymer networks. J. Mech. Phys. Solids **191**, 105732 (2024). https://doi.org/10.1016/j.jmps.2024.105732
89. X. Yu, S. Zhou, X. Zheng, T. Guo, Y. Xiao, B. Song, A biodegradable shape-memory nanocomposite with excellent magnetism sensitivity. Nanotechnology **20**(23), 235702 (2009). https://doi.org/10.1088/0957-4484/20/23/235702
90. A. Ahangari, H. Doostmohammadi, M. Baniassadi, M. Bodaghi, M. Baghani, Development and 4D printing of magneto-responsive PMMA/TPU/Fe3O4 nanocomposites with superior shape memory and toughness properties. Eur. Polym. J. **220**, 113495 (2024). https://doi.org/10.1016/j.eurpolymj.2024.113495
91. P.R. Buckley et al., Inductively heated shape memory polymer for the magnetic actuation of medical devices. IEEE Trans. Biomed. Eng. **53**(10), 2075–2083 (2006). https://doi.org/10.1109/TBME.2006.877113
92. R.U. Hassan, S. Jo, J. Seok, Fabrication of a functionally graded and magnetically responsive shape memory polymer using a 3D printing technique and its characterization. J. Appl. Polym. Sci. **135**(11), 45997 (2018). https://doi.org/10.1002/app.45997
93. M. Lalegani Dezaki, M. Bodaghi, Sustainable 4D printing of magneto-electroactive shape memory polymer composites. Int. J. Adv. Manufact. Technol. **126**(1), 35–48 (2023). https://doi.org/10.1007/s00170-023-11101-0
94. P. Srinivas et al., Mechanical properties, energy absorption, and shape memory behavior of 3D printed PLA-MXene nanocomposites and gyroid lattices. Adv. Eng. Mater. **26**(12), 2301698 (2024). https://doi.org/10.1002/adem.202301698
95. I. Abdollahi, R. Sedaghati, Investigation of macroscopic mechanical behavior of magnetorheological elastomers under shear deformation using microscale representative volume element approach. Polymers (Basel) **16**(10) (2024). https://doi.org/10.3390/polym16101374

96. J. Berasategi et al., Anisotropic behaviour analysis of silicone/carbonyl iron particles magnetorheological elastomers. Rheol. Acta **59**(7), 469–476 (2020). https://doi.org/10.1007/s00397-020-01218-4
97. L. Costi, A. Georgopoulou, S. Mondal, F. Iida, F. Clemens, 3D printable self-sensing magnetorheological elastomer. Macromol. Mater. Eng. **309**(2), 2300294 (2024). https://doi.org/10.1002/mame.202300294
98. Z. Peng, Z. Zhai, R. Yang, H. Xu, Y. Wu, Fabrication and property research of a new 3D-Printable magnetorheological elastomer (MRE). Mater. Today Phys. **45**, 101467 (2024). https://doi.org/10.1016/j.mtphys.2024.101467
99. J.C. Breger et al., Self-folding thermo-magnetically responsive soft microgrippers. ACS Appl. Mater. Interfaces **7**(5), 3398–3405 (2015). https://doi.org/10.1021/am508621s
100. H. Haider et al., Exceptionally tough and notch-insensitive magnetic hydrogels. Soft Matter. **11**(42), 8253–8261 (2015). https://doi.org/10.1039/C5SM01487E
101. L. Kong et al., Self-adaptive magnetic photonic nanochain cilia arrays. Adv. Funct. Mater. **30**(52), 2005243 (2020). https://doi.org/10.1002/adfm.202005243
102. Z. Wang et al., Magnetic helical hydrogel motor for directing t cell chemotaxis. Adv. Funct. Mater. **31**(25), 2101648 (2021). https://doi.org/10.1002/adfm.202101648